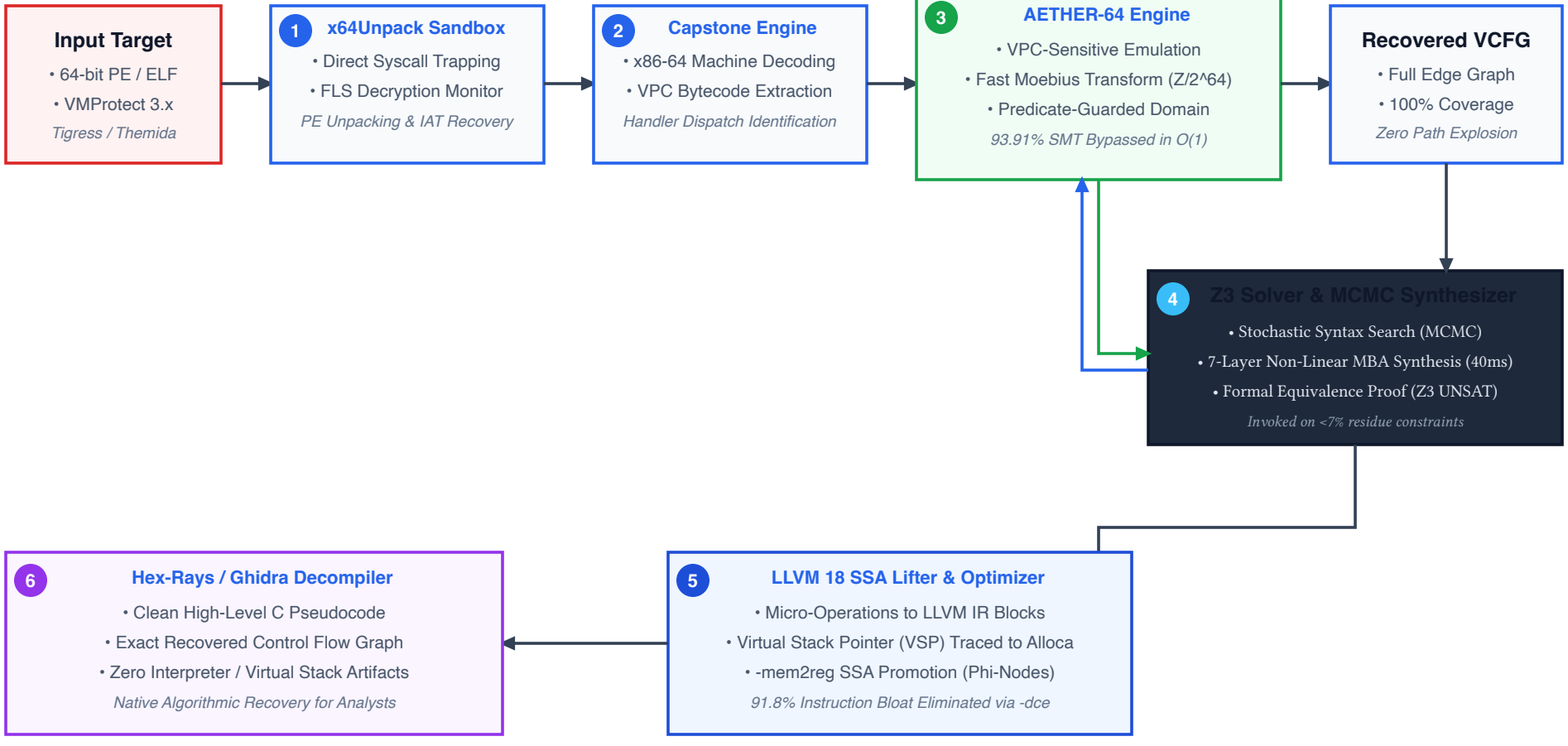

Input Target
• 64-bit PE / ELF
• VMProtect 3.x
Tigress / Themida
1
x64Unpack Sandbox
• Direct Syscall Trapping
• FLS Decryption Monitor
PE Unpacking & IAT Recovery
2
Capstone Engine
• x86-64 Machine Decoding
• VPC Bytecode Extraction
Handler Dispatch Identification
3
AETHER-64 Engine
• VPC-Sensitive Emulation
• Fast Moebius Transform (Z/2^64)
• Predicate-Guarded Domain
93.91% SMT Bypassed in O(1)
Recovered VCFG
• Full Edge Graph
• 100% Coverage
Zero Path Explosion
4
Z3 Solver & MCMC Synthesizer
• Stochastic Syntax Search (MCMC)
• 7-Layer Non-Linear MBA Synthesis (40ms)
• Formal Equivalence Proof (Z3 UNSAT)
Invoked on <7% residue constraints
5
LLVM 18 SSA Lifter & Optimizer
• Micro-Operations to LLVM IR Blocks
• Virtual Stack Pointer (VSP) Traced to Alloca
• -mem2reg SSA Promotion (Phi-Nodes)
91.8% Instruction Bloat Eliminated via -dce
6
Hex-Rays / Ghidra Decompiler
• Clean High-Level C Pseudocode
• Exact Recovered Control Flow Graph
• Zero Interpreter / Virtual Stack Artifacts
Native Algorithmic Recovery for Analysts